

# AI Capstone Final Project Report - Advance

## Group 13

### I. Task Description

This project studies imitation-learning-based robotic manipulation in a Physical AI pipeline. The Advanced-level post-meal tabletop cleanup task, which is implemented and evaluated in simulation using the course-provided Isaac Sim / Isaac Lab and LeRobot-based framework.

#### A. Task: Dining cleanup

For the Advanced-level extension, we designed a new task aimed to tackle **Post-Meal Tabletop Cleanup with Tableware Clearing and Surface Wiping**. This task extends the Entry-level cutlery arrangement problem from simple object placement to a more complete household cleanup scenario.

The Advanced-level scene contains a dining table, a single robot arm, target tableware objects, a tray, a wiping cloth or simplified wiping tool, protected non-target objects, and a predefined dirty region. The robot is expected to perform the following sequence:

1. Identify and move target tableware objects into the tray.
2. Retrieve or use the cloth.
3. Wipe the predefined dirty region on the table.
4. Avoid disturbing protected non-target objects.
5. Avoid dropping objects or pushing objects off the table.
6. Complete the task within a fixed episode horizon.

The Advanced-level task introduces several additional challenges. First, it is a **multi-stage manipulation task** rather than a single pick-and-place objective. Second, the robot must distinguish between object roles: target tableware, receptacle, tool, protected object, and dirty region. Third, the task includes both **discrete object manipulation and continuous surface-coverage** behavior. Finally, the robot must avoid unsafe interactions with **protected objects**, which makes the task a constrained manipulation problem.

The motivation for this Advanced-level task is that post-meal cleanup is a common and repetitive household activity. A robot capable of clearing tableware and wiping a table surface could be useful in home assistance, elder care, mobility assistance, and service robotics. The proposed task is therefore designed to test whether an imitation-learning policy can perform role-aware, sequential, and safety-constrained tabletop manipulation.

## II. Data Collection

### A. Advance-Level Simulation Data Collection

For the Advanced-level task, we designed a new dining cleanup task named HCIS-DiningCleanup-SingleArm-v0. The robot is required to clear target tableware into a tray and perform a simplified wiping behavior while avoiding protected non-target objects.

The Advanced-level task is conducted entirely in simulation. We prepared a new task configuration and object assets. The scene contains target tableware objects, a tray, a wiping cloth or simplified wiping tool, protected objects, and a predefined dirty region.

The Advanced-level training dataset contains 77 successful demonstrations generated by the FSM planner from 100 provided object poses. Both ACT and Diffusion Policy were trained using the same 77-demonstration dataset. This makes the Advanced-level policy comparison fair because the training data is controlled and the main difference is the policy architecture.

### B. Advance-Level Data Generation FSM Design

For the Advanced-level task, we designed a new FSM for the dining cleanup environment. The Advanced-level task requires multi-stage manipulation. The robot must clear tableware into a tray, use a cloth to wipe a targeted dirty region, and avoid disturbing protected objects.

The Advanced-level FSM contains the following three main stages:

The first stage is tableware clearing. The robot sequentially manipulates the bowl and the spoon. For each object, the FSM moves above the object, approaches the grasp pose, closes the gripper, lifts the object, moves above the tray, lowers to the release pose, and retreats. Object-specific grasp and drop offsets are used because the bowl and spoon have different shapes and require different grasping and placement strategies. For example, the bowl uses an edge-biased grasp target, while the spoon uses a smaller grasp height and a different drop offset in the tray.

The second stage is cloth acquisition. After the tableware objects are cleared, the robot moves to the cloth, approaches it, grasps it, and lifts it. This stage prepares the robot for the wiping behavior.

The third stage is table wiping. The FSM moves the cloth above the wiping region, lowers it to the contact height, and performs multiple sweeping lanes across the predefined dirty region. The wiping lanes are defined over the left table region and are ordered from right to left to reduce the risk of pushing objects off the table. During wiping, the FSM tracks the cloth footprint over a

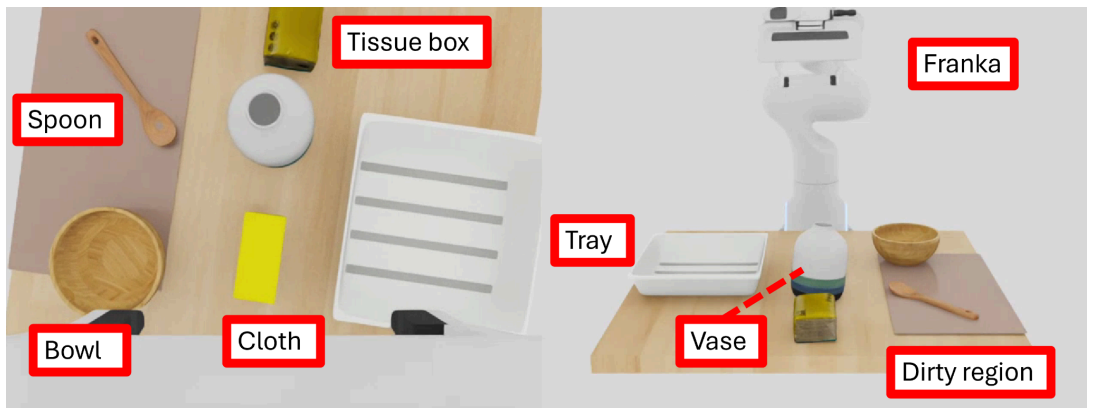
discretized grid and updates the covered cells when the cloth is within the valid contact-height range.

The Advanced-level FSM also implements a detailed success-checking function. The rollout is considered successful only if all required conditions are satisfied:

1. The bowl is placed inside the tray.
2. The spoon is placed inside the tray.
3. The cloth covers enough of the predefined wiping region.
4. The cloth is returned to the tray after wiping.
5. The tray remains close to its initial position.
6. Protected objects, such as the tissue box and vase, remain within the allowed displacement tolerance.
7. No non-tableware object falls off the table to its initial position.
8. The complete FSM timeline is finished.

This success definition is strict because the Advanced-level task is multi-stage and safety-constrained. A rollout should not be counted as successful if it clears tableware but fails to wipe the table, or if it wipes the table while disturbing protected objects.

The Advanced-level FSM is not only a data-generation tool but also part of the task definition. It defines the intended expert behavior, the task sequence, and the evaluation criteria. By decomposing the cleanup task into clearing and wiping stages, the FSM makes the long-horizon task feasible to reproduce and analyze. At the same time, the evaluation variants test whether the trained policy can generalize beyond the base FSM-generated training distribution.



**Figure 1. Fixed spoon direction evaluation scene**

### III. Policy Training

#### A. Model Architecture

We trained two imitation-learning policies, ACT and Diffusion Policies, and one visuomotor policy, VLA.

## **Action Chunking with Transformers, ACT**

Action Chunking with Transformers predicts a chunk of future robot actions from the current visual observation and robot state. This architecture is suitable for long-horizon manipulation because it models temporally extended behaviors such as approaching, grasping, lifting, transferring, and releasing objects.

During inference, ACT uses temporal ensembles to combine overlapping action predictions. This helps smooth the predicted action sequence and reduces abrupt changes between consecutive actions. This property is useful for the Advanced-level task because stable grasping and accurate placement require temporally consistent control.

We used ACT as the training policy because the Advanced-level task is a long-horizon and multi-step manipulation task. The policy must maintain temporal consistency across approach, grasp, lift, transfer, and release phases. ACT is suitable for this setting because its chunk-based action prediction can model multi-step manipulation primitives more naturally than purely single-step behavior cloning.

## **Diffusion Policy**

Diffusion Policy formulates visuomotor action prediction as a conditional denoising process. Given visual observations and robot states, the policy generates action trajectories through iterative denoising. Compared with deterministic behavior cloning, Diffusion Policy can better represent multimodal action distributions and generate smooth action sequences.

In this project, Diffusion Policy was trained as a comparison against ACT. ACT and Diffusion Policy were trained on the same 77 FSM-generated demonstrations.

## **SmolVLA**

Recently, the VLA has become more and more popular in the field of robot manipulation. We also try to train SmolVLA as an additional VLA-based policy. Unlike ACT and Diffusion Policy, which mainly learn visuomotor control from image and state observations, SmolVLA follows a vision-language-action policy design. We included this experiment to explore whether a VLA-style policy can also learn the manipulation behavior from our generated demonstration dataset.

For the Advanced-level task, SmolVLA was trained using a larger 300-demonstration dataset. Since the Advanced-level SmolVLA experiment

uses a different data size from ACT and Diffusion Policy, we treat it as an additional exploratory experiment rather than a strictly controlled comparison.

#### B. Advanced-Level Training Procedure

For advanced-level experiments, we trained ACT, Diffusion Policy, and SmolVLA using the LeRobot training pipeline. The overall procedure was:

1. Generate the simulation dataset in LeRobot format.
2. Load the dataset from the local path or Hugging Face LeRobot cache.
3. Launch `lerobot-train` with the selected ACT configuration.
4. Enable image transforms for visual robustness.
5. Train the policy for the configured number of steps.
6. Save checkpoints periodically.
7. Upload or copy the final `pretrained_model` checkpoint.
8. Run local rollout evaluation in Isaac Sim.

For the Advanced-level task, both ACT and Diffusion Policy were trained using the same 77 FSM-generated demonstrations. This dataset was generated from the custom dining cleanup simulation environment. We additionally trained SmolVLA on the same Advanced-level dataset with 77 demonstrations.

The Advanced-level training setup is summarized below.

| Policy           | Training dataset                    | Number of demonstrations |
|------------------|-------------------------------------|--------------------------|
| ACT              | Advanced dining cleanup FSM dataset | 77                       |
| Diffusion Policy | Advanced dining cleanup FSM dataset | 77                       |
| SmolVLA          | Advanced dining cleanup FSM dataset | 77                       |

**Table 1. Summary of Advanced-level training setup**

For ACT, chunk size was selected to be 100 since each FSM demonstration stage is roughly around 100 steps, allowing the model to learn a complete manipulation phase such as approach, grasp, lift, or placement. We used `n_action_steps = 1` so the policy replans after every executed action, which improves closed-loop reactivity for thin-object grasping. And `temporal_ensemble_coeff = 0.01` was used to smooth overlapping action predictions and reduce abrupt motion.

| Key ACT hyperparameters        | Numerical Value |
|--------------------------------|-----------------|
| policy.chunk_size              | 100             |
| policy.n_action_steps          | 1               |
| policy.temporal_ensemble_coeff | 0.01            |

**Table 2. Key ACT hyperparameters**

For Diffusion policy, horizon = 16 was used to predict a short future action trajectory through the denoising model. We used n\_action\_steps = 8 so that multiple predicted actions are executed before replanning, producing smoother motion while still allowing periodic correction. n\_obs\_steps = 2 gives the model short-term temporal context, which helps identify whether the robot is approaching, grasping, or transporting an object. The diffusion model used 100 DDPM training timesteps with epsilon prediction. The optimizer settings were kept the same as ACT.

| Key Diffusion Policy hyperparameters | Numerical Value |
|--------------------------------------|-----------------|
| horizon                              | 16              |
| n_action_steps                       | 8               |
| n_obs_steps                          | 2               |

**Table 3. Key diffusion policy hyperparameters**

### C. Advanced-Level Training Procedure

For the Advanced-level task, both ACT and Diffusion Policy were trained using the same 77 FSM-generated demonstrations. This dataset was generated from the custom dining cleanup simulation environment. We additionally trained SmolVLA on the same Advanced-level dataset with 77 demonstrations. The Advanced-level training setup is summarized below.

| Policy           | Training dataset                    | Number of demonstrations |
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| ACT              | Advanced dining cleanup FSM dataset | 77                       |
| Diffusion Policy | Advanced dining cleanup FSM dataset | 77                       |

|         |                                        |    |
|---------|----------------------------------------|----|
| SmolVLA | Advanced dining<br>cleanup FSM dataset | 77 |
|---------|----------------------------------------|----|

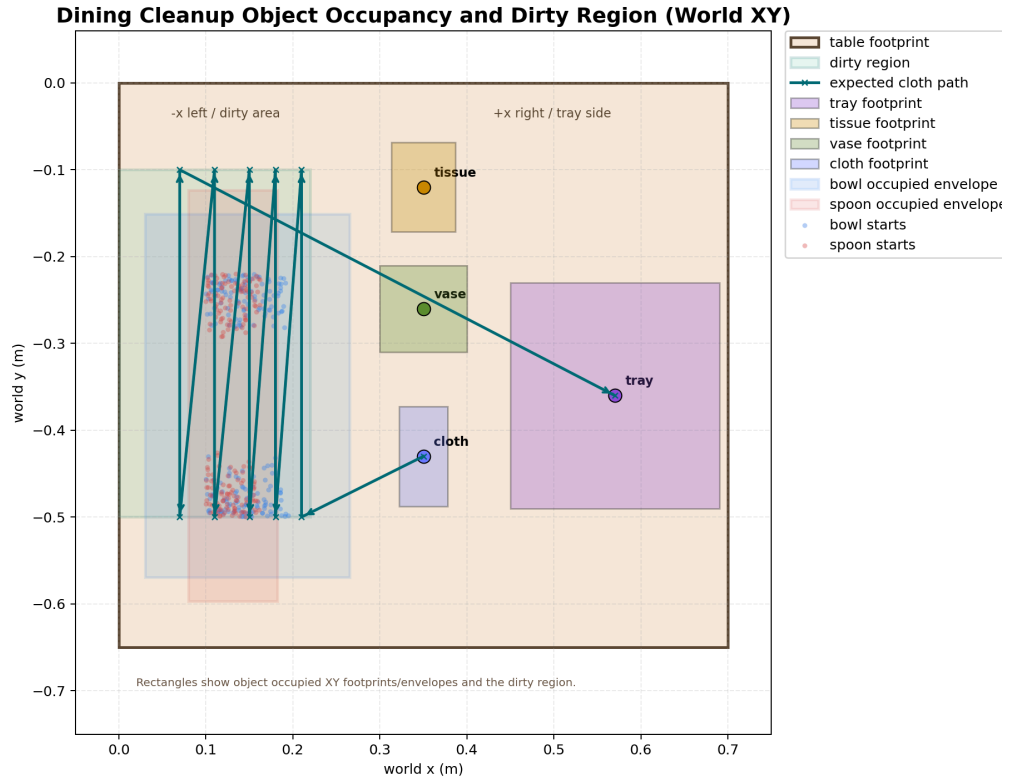
**Table 4. Summary of Advanced-level training setup**

#### IV. Experimental Results

Our experiments were designed to answer **whether the trained policy can generalize across different environment configurations**, including fixed spoon direction, random spoon direction, and changed USD assets.

##### A. Data Quality Analysis

In the data quality evaluation, object randomization is used as the primary metric. For Advanced settings, object poses are sampled uniformly within a predefined valid range. The underlying assumption is that more uniformly randomized object positions encourage greater diversity in FSM behavior, which in turn leads to policies with improved generalization capability.



**Figure 2. Advance-level object 100-pose distribution.**

##### B. Evaluation method

In our task, the evaluation consists of multiple sequential objectives that follow the intended execution order of the robot trajectory.

First, the robot is required to place the bowl and spoon into the tray. Success is determined by measuring the distance between each object and the tray after placement. If the objects are located within a predefined threshold, the objective is considered successful.

Next, the Franka robot grasps a cleaning cloth and wipes the tabletop. To evaluate this stage, we calculate the coverage area of the wiping trajectory. A threshold of 0.7 is used, meaning that the wiped area must exceed 70% of the ideal target cleaning area for the wiping task to be considered successful.

Finally, the robot must return the cleaning cloth to the tray. Similar to the bowl and spoon placement task, success is determined based on the final distance between the cloth and the tray.

Another important aspect of the evaluation is safety and environmental preservation. The robot should complete the task without disturbing other objects on the table. Therefore, we additionally monitor several static objects, including the tray, vase, and tissue box, and detect whether they have moved relative to their initial positions. Any significant displacement of these objects is treated as an undesirable interaction and is recorded as a task violation.

```
[Evaluation] Episode 2 Stage Status
Tableware : success
  Bowl XY   : success
  Bowl dx/dy: 0.001 / 0.047
  Spoon XY  : success
  Spoon dx/dy: 0.042 / 0.065

Wiping     : success
  Cloth XY  : success
  Cloth dx/dy: 0.129 / 0.016
  Cloth z/contact: 0.151 / fail
  Coverage  : success
  Ratio     : 0.886
  Threshold : 0.565
  Ideal     : 0.807
  Required  : 70%

Tray       : dist=0.059
Protected  : success
  Tissue    : success
  Tissue dist: 0.001
  Vase      : success
  Vase dist : 0.000
  Min non-tableware z: 0.041

Overall    : success
```

**Figure 3. Evaluation log format**

To measure the wiping coverage area, we conducted a detailed analysis of the evaluation pipeline and found that the rollout process executes the termination-condition checks after every simulation step. Based on this



observation, we incorporated the area accumulation logic into the `dining_cleanup_success` function, which is responsible for determining task success.

During evaluation, the wiped area is accumulated incrementally throughout the rollout. To ensure that only valid wiping motions contribute to the coverage metric, we additionally consider the contact height between the cleaning cloth and the tabletop. Specifically, a region is counted as wiped only when the vertical distance between the cloth and the table surface is less than 4 cm. This constraint prevents motions performed above the table from being incorrectly classified as successful wiping behavior and provides a more accurate estimate of the effective cleaning coverage.

```
coverage_ratio = _update_wipe_coverage_ratio(
    env,
    cloth_pos,
    wipe_x_range=wipe_x_range,
    wipe_y_range=wipe_y_range,
    cloth_xy_size=cloth_xy_size,
    resolution=wipe_coverage_resolution,
    contact_z_range=wipe_contact_z_range,
)
```

```
half_x = 0.5 * cloth_xy_size[0]
half_y = 0.5 * cloth_xy_size[1]
in_contact = torch.logical_and(
    cloth_pos[:, 2] >= contact_z_range[0],
    cloth_pos[:, 2] <= contact_z_range[1],
)
covered_now = torch.logical_and(
    torch.abs(cloth_pos[:, 0, None, None] - grid_x[None, :, None]) <= half_x,
    torch.abs(cloth_pos[:, 1, None, None] - grid_y[None, None, :]) <= half_y,
)
state |= torch.logical_and(covered_now, in_contact[:, None, None])
return state.float().mean(dim=(1, 2))
```

**Figure 4. 5. Mesh update logic**

### C. Different Environment Configurations

For the Advanced-level task, we evaluate the generalization capability of the trained policy across different environment configurations. Upon training completion, rollout evaluations are conducted under three scenarios: **fixed spoon direction**, **random spoon direction**, and **changed USD assets to** assess the robustness of the ACT and Diffusion Policy models.

1. **Fixed spoon direction:** This setting corresponds to the original training environment. During data collection, the spoon is always

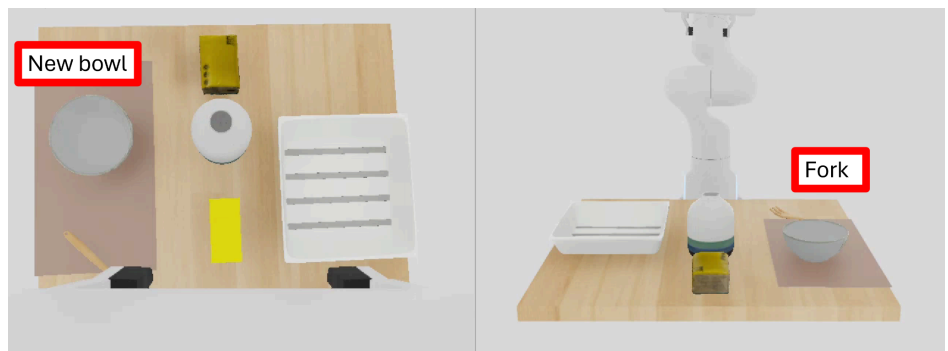
placed at a fixed orientation of 45 degrees, and the policy is evaluated under the same configuration used during training.

2. **Random spoon direction:** To assess whether the policy can generalize beyond the training distribution, we randomize the spoon orientation during evaluation. Since all training demonstrations were collected with the spoon fixed at 45 degrees, this experiment tests whether the policy can adapt its end-effector trajectory and successfully grasp spoons presented at previously unseen orientations.



**Figure 6. Random spoon direction evaluation scene**

3. **Changed USD assets:** This scenario introduces a more substantial distribution shift. In real-world dining environments, tableware is unlikely to remain identical across deployments. Therefore, we investigate whether the policy can generalize based on the geometric characteristics of objects rather than their specific visual appearance. To keep the task feasible, we replace the original assets with objects that require similar manipulation strategies, including bowls with different appearances and forks that share a similar shape with the original spoon. This setup allows us to evaluate whether the policy can transfer its learned grasping behavior to visually different but functionally similar objects.

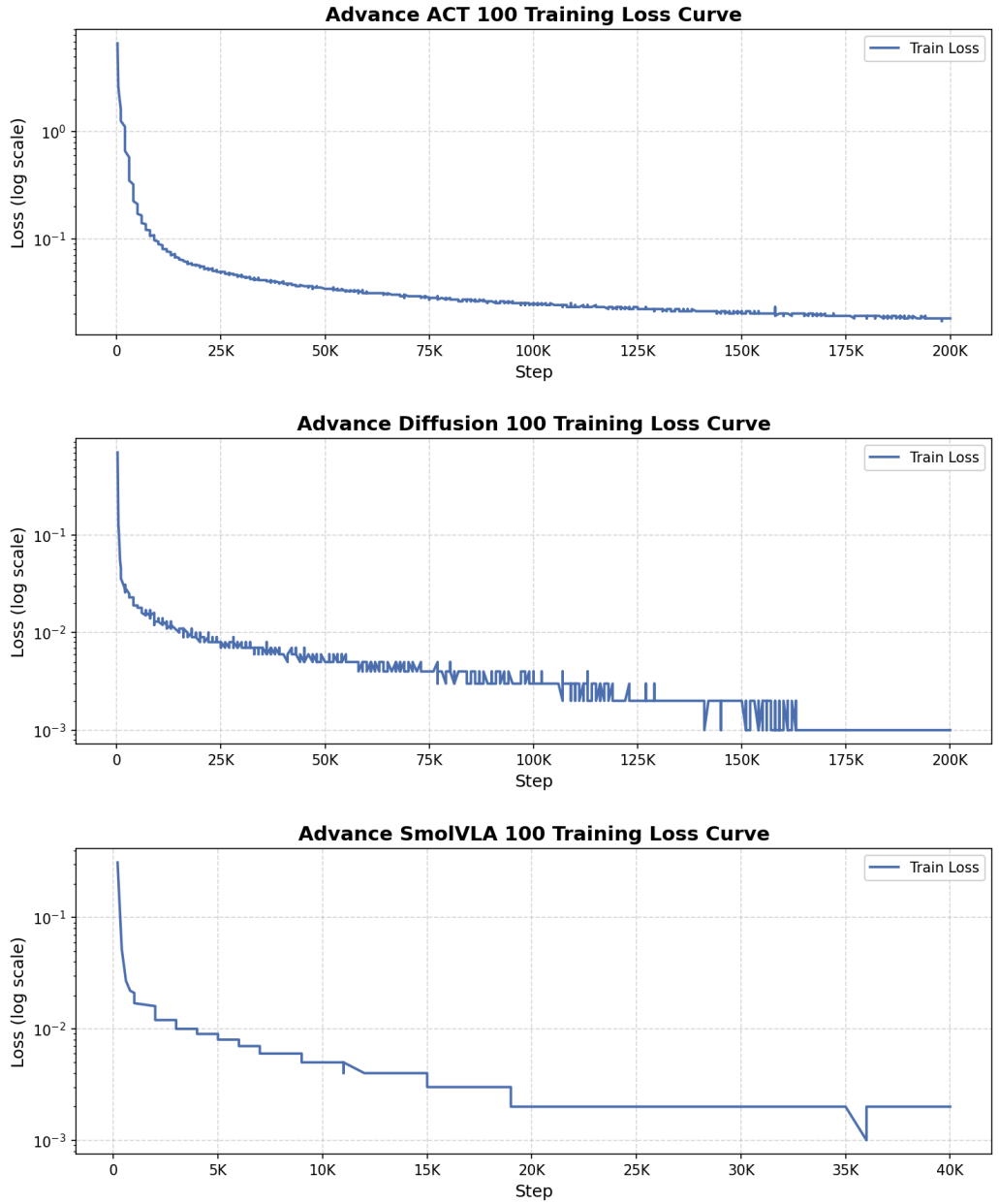


**Figure 7. Changed USD asset evaluation scene**

#### D. Training Loss

The loss curves of all three models exhibit trends that are largely consistent with those observed in the Entry-level task. This suggests that the trajectories generated by the FSM are relatively easy for the policies to learn, allowing all models to achieve low training loss and stable convergence.

In the Entry-level experiments, we trained all models for 100k steps to enable a fair comparison across architectures. However, the Advanced-level task involves significantly longer demonstrations, as each FSM execution contains substantially more timesteps. To account for the increased trajectory length and provide sufficient training exposure, we increased the training budget to 200k steps for all Advanced-level experiments.



## E. Eval results

For the evaluation, we focus on two main comparisons. First, we aim to assess the performance of different policy architectures on the task. To ensure a fair comparison, all models are trained and evaluated using the same dataset, allowing us to isolate the effect of the policy architecture itself.

Second, we evaluate the generalization capability of a single ACT policy across different environment configurations. By testing the same trained policy under varying conditions, including unseen object orientations and modified assets, we can examine how well the learned behavior transfers beyond the training distribution and whether the policy remains robust under environmental changes.

### 1) compare different policy type

We evaluated three model architectures—ACT, Diffusion Policy, and SmolVLA—following the same setup as in the Entry-level experiments. However, the results in the Advanced-level task show a reversed performance ranking compared to the Entry setting.

SmolVLA performs the worst among the three models. It fails to demonstrate meaningful task execution, and in most rollouts, errors occur immediately at the first step, i.e., the bowl grasping stage. The gripper is often positioned incorrectly and fails to make contact with the bowl. Moreover, unlike its behavior in the Entry-level task, it does not exhibit any corrective behavior after failure. Instead, the policy becomes stuck in repetitive oscillatory motions until episode timeout.

In contrast, ACT achieves the best overall performance, with a success rate of 76%. Failure cases can be categorized into several types. First, during bowl grasping, insufficient insertion depth leads to unstable grasps at the edge of the bowl, causing the object to slip during lifting. Second, during cloth grasping, the Franka arm sometimes exhibits abnormal oscillatory motion, which in severe cases leads to collisions with the tray and causes other objects to be displaced. Third, in rare cases, after successfully grasping both the bowl and cloth, the wiping behavior degenerates into a fixed back-and-forth motion without covering sufficient area, resulting in timeout due to insufficient cleaning coverage.

Diffusion Policy performs less favorably. It achieves a 66% success rate in the bowl grasping stage, but only 20% in spoon grasping. From rollout observations, the gripper frequently targets incorrect spatial locations when attempting to grasp the spoon, often closing in empty

space. After failure in the spoon stage, the subsequent behavior becomes unstable due to out-of-distribution states not seen during training. As a result, the diffusion-based policy exhibits highly unpredictable trajectories and a significant drop in downstream task success. Additionally, compared to ACT and SmolVLA, Diffusion Policy shows more stochastic and less controlled motion patterns. This is reflected in the no-collision statistics, where only the diffusion model exhibits collisions with other objects on the table, indicating weaker motion consistency and spatial awareness.

We attribute these performance differences to the structure of the Advanced-level FSM trajectories. In this setting, the demonstrations are significantly longer and highly repetitive. The primary source of stochasticity lies only in the initial positions of the bowl and spoon. Once the object locations are correctly identified, the subsequent execution follows a largely deterministic and repeated sequence of actions.

This structure has different implications for each model. For ACT, which is fundamentally a behavior cloning model, such regularity is beneficial. As long as the model captures the underlying trajectory pattern, it can reproduce the full sequence reliably, leading to strong performance in the Advanced task.

In contrast, SmolVLA appears less capable of leveraging this structured multi-stage procedure in our setup. Since no language prompts are provided and the model is trained purely from trajectory data, it may lack an explicit representation of task stages or hierarchical structure. As a result, it struggles to decompose the task into sequential steps, which leads to early failures and the absence of recovery behaviors. This is particularly evident compared to the Entry-level task, where the objective is simpler and more direct—placing utensils into a target location. Although the Entry setting involves a wider spatial distribution of object placements, SmolVLA performs better there, suggesting that it benefits more from simpler goal structures even under higher spatial variability.

|                               | ACT    | Diffusion | SmolVLA |
|-------------------------------|--------|-----------|---------|
| Overall success               | 76.7%  | 0.0%      | 0.0%    |
| Bowl placement                | 86.7%  | 66.7%     | 0.0%    |
| Spoon placement               | 86.7%  | 20.0%     | 0.0%    |
| Wipe coverage (avg)           | 52.1%  | 5.7%      | 0.0%    |
| Coverage pass ( $\geq 70\%$ ) | 76.7%  | 16.7%     | 0.0%    |
| Cloth returned                | 76.7%  | 6.7%      | 0.0%    |
| No collision                  | 100.0% | 60.0%     | 100.0%  |

**Table 5. Success rates details for different policy**

2) apply policy to unseen scene

We evaluate three scenarios, with results summarized in the figure. The original setting has been discussed previously, so the focus here is on the random spoon direction and changed USD assets conditions.

In the random spoon direction setting, the overall success rate drops significantly. The bowl grasping performance remains consistent with the original configuration, indicating that the degradation mainly comes from the spoon grasping stage, where success decreases by approximately 30%. From rollout observations, the gripper orientation does not actively adapt to the spoon’s rotation. As a result, successful grasping largely depends on whether the randomly oriented spoon happens to align with the fixed gripper approach direction, making the task highly sensitive to orientation mismatch.

In the changed USD assets setting, failures occur more frequently during bowl grasping. This is primarily due to the substituted objects having more curved bowl geometries, which are less compatible with the Franka gripper and reduce grasp stability. Although visual localization remains reliable, physical interaction becomes less robust, leading to frequent grasp failures. In contrast, the fork and spoon replacements share similar geometric properties, allowing a reasonable number of successful grasps. However, the overall success rate remains at 44% because once bowl grasping fails, the policy often diverges into unstable and unpredictable behaviors, which subsequently affects the remaining task stages.

|                               | Original | Random spoon direction | Changed USD assets |
|-------------------------------|----------|------------------------|--------------------|
| Overall success               | 76.7%    | 43.3%                  | 3.3%               |
| Bowl placement                | 86.7%    | 86.7%                  | 50.0%              |
| Spoon placement               | 86.7%    | 50.0%                  | 44.4%              |
| Wipe coverage (avg)           | 52.1%    | 26.9%                  | 14.7%              |
| Coverage pass ( $\geq 70\%$ ) | 76.7%    | 50.0%                  | 5.6%               |
| Cloth returned                | 76.7%    | 46.7%                  | 5.6%               |
| No collision                  | 100.0%   | 90.0%                  | 100.0%             |

**Table 5. Success rates details for different configs**

## V. Discussion

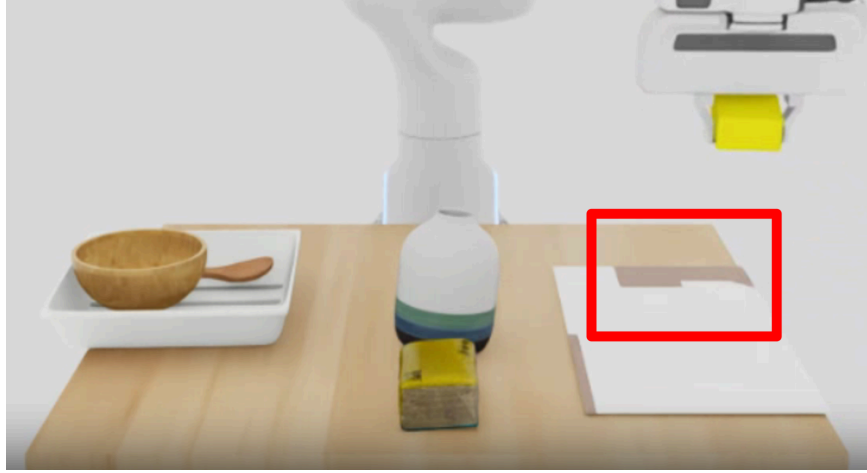
A. We observed that there is consistently a small region that cannot be wiped during the cleaning task.

1) FSM-related issue:

Although the FSM specifies a straight-line trajectory between two points for the wiping motion, the Franka robot often holds the cloth at a tilted angle during execution. Attempts to reduce the friction between the cloth and the table did not resolve the issue. We later identified that the root cause was the wiping height defined in the FSM being too low, which caused the cloth to penetrate into the table surface, leading to an unrealistic interaction between simulation and physical behavior.

2) Suspected robot joint limitation:

We further hypothesize that the issue is also related to Franka’s joint constraints. We repeatedly adjusted the FSM-defined wiping trajectory. Initially, the starting point was set at the top-right corner of the dirty area. However, the robot exhibited very slow and unstable motion, sometimes even extending toward outside the table boundary. We then changed the starting point to the top-left corner. Even after this adjustment, a small region in the top-right corner still consistently remained uncleaned, despite being covered in the planned trajectory. We attribute this persistent failure to the limitations of Franka’s joint configuration and its reachable workspace.



**Figure 8. The small region that cannot be wiped**

- 3) In addition, we initially expected the table-wiping stage to be relatively easy for the policy to learn because the wiping motion follows a largely fixed trajectory across demonstrations. Since the behavior is highly repetitive and exhibits little variation, we assumed that the policy would be able to reproduce it consistently.

However, the experimental results revealed that this was not the case. Despite the repetitive nature of the wiping trajectory, the policy's behavior during this stage remained noticeably inconsistent. Successful wiping was not guaranteed in every rollout, and the robot occasionally deviated from the intended trajectory or failed to achieve sufficient coverage. This observation suggests that accurately reproducing a long-horizon manipulation task involves more than simply memorizing motion patterns. Small errors accumulated during earlier stages, variations in object states, and control inaccuracies can all affect the execution of subsequent actions, making the wiping stage more challenging than initially anticipated.

## B. Stain

The design of the stain generation mechanism is intended to encourage the policy to fully wipe the entire surface. This design is inspired by a common daily behavior: even when a table does not appear visibly dirty, it is still typically wiped across the entire surface after a meal. The underlying assumption is that if the policy can reliably perform full-area coverage, then even partially unobserved dirty regions will also be cleaned during execution.

For implementation, we construct a mesh to represent the stain region and update it at each simulation step. The stain tracking and area computation are performed simultaneously during rollout. If a mesh cell has been wiped, it is



marked as clean (shown in white); otherwise, it remains in the stained state (shown in brown). This representation allows us to dynamically track the cleaning progress over time and directly compute the wiped coverage, which is then used to evaluate whether the task meets the required cleaning threshold.

### C. Future work

Our work is motivated by a practical everyday scenario, and deployment in real-world environments is the ultimate objective. However, data collected in simulation inevitably omits several challenges that would arise in real applications. For example, bowls and spoons used during a meal are often covered with food residue or grease, which may affect grasp stability and manipulation performance. Such factors were not modeled in our simulation environment and therefore are not reflected in the training data.

Another limitation concerns the cleaning cloth. Simulating deformable objects remains challenging in many robotics simulators, so in this project the cloth was approximated as a rigid cuboid. While this simplification makes data generation and training more tractable, a realistic deployment would require a deformable cloth model that better captures the physical properties of fabric. This would likely require further refinement of the simulator assets and physics configuration.

Finally, our policies show a noticeable performance degradation when the tableware assets are replaced. This suggests that the learned policies are not yet robust to the wide variety of object appearances and geometries encountered in everyday environments. Future work should therefore focus on improving both visual generalization and grasp adaptation, enabling the robot to recognize unfamiliar objects and automatically adjust its manipulation strategy according to their shape and physical characteristics.